

PRAKTIKUM SISTEM OPERASI

MODUL V

(EKSPLORASI PROSES)



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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JURNAL

1. Jawablah pertanyaan berikut ini:

a. Berapa banyaknya maksimum proses yang ada pada Xinu?

Jawab:

Jumlah maksimum proses pada Xinu ditentukan oleh konstanta NPROC yang ada di file process.h.

Maksimum proses pada Xinu adalah 50 proses.

b. Berapa maksimal panjang nama suatu proses pada Xinu?

Jawab:

Panjang nama proses diatur oleh konstanta PNMLEN pada process.h.

Maksimal panjang nama proses adalah 16 karakter (termasuk null-terminator).

c. Berapa nilai prioritas awal pada saat proses dibuat?

Jawab:

Saat proses dibuat menggunakan fungsi create(), nilai prioritas biasanya diberikan sebagai parameter saat pemanggilan. Nilai prioritas awal ditentukan saat proses dibuat (parameter pada fungsi create), namun secara umum default yang sering digunakan adalah 20.

d. Ada berapa total state pada Xinu? Sebutkan!

Jawab:

state proses dapat dilihat di process.h.

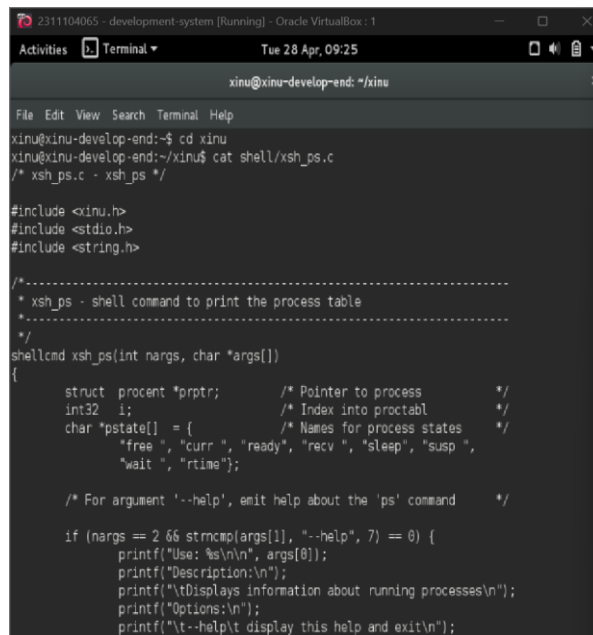
Total terdapat 7 state, yaitu:

1. PR_FREE → proses kosong / tidak digunakan
2. PR_CURR → proses sedang berjalan
3. PR_READY → proses siap dijalankan
4. PR_RECV → proses sedang menunggu pesan
5. PR_SLEEP → proses sedang tidur
6. PR_SUSP → proses ditangguhkan
7. PR_WAIT → proses menunggu resource

2. Perintah `ps` adalah perintah untuk menampilkan statistik process yang berjalan. Source code dari `ps` tersimpan pada file `xsh_ps.c`. Carilah file tersebut dan beri komentar pada 20 baris terakhir di source code tersebut!

Jawab:

- Cari file `xsh_ps.c`



```
2311104065 - development-system [Running] - Oracle VM VirtualBox : 1
Activities Terminal Tue 28 Apr, 09:25
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ cd xinu
xinu@xinu-develop-end:~/xinu$ cat shell/xsh_ps.c
/* xsh_ps.c - xsh_ps */

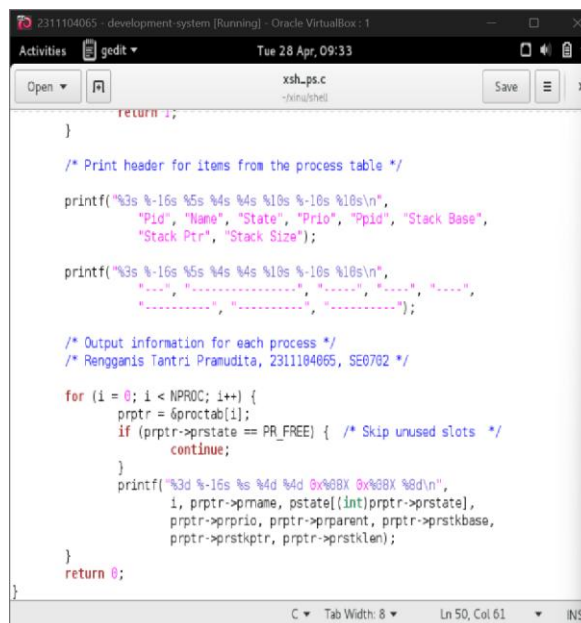
#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*
 * xsh_ps - shell command to print the process table
 */
shellcmd xsh_ps(int nargs, char *args[])
{
    struct procent *prptr; /* Pointer to process */
    int32 i; /* Index into proctabl */
    char *pstate[] = { /* Names for process states */
        "free ", "curr ", "ready", "recv ", "sleep", "susp ",
        "wait ", "rtimn"};

    /* For argument '--help', emit help about the 'ps' command */

    if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
        printf("Use: %s\n\n", args[0]);
        printf("Description:\n");
        printf("\tDisplays information about running processes\n");
        printf("Options:\n");
        printf("\t-t--helpit display this help and exit\n");
    }
}
```

- Menambahkan komentar nama, nim, dan kelas



```
xsh_ps.c
~(xinu/shell)

return ;
}

/* Print header for items from the process table */

printf("%3s %16s %5s %4s %4s %10s %10s %10s\n",
    "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
    "Stack Ptr", "Stack Size");

printf("%3s %16s %5s %4s %4s %10s %10s %10s\n",
    "----", "-----", "-----", "-----", "-----",
    "-----", "-----");

/* Output information for each process */
/* Rengganis Tantri Pramudita, 2311104065, SE0702 */

for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->pstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %16s %5s %4d %4d 0x%08X 0x%08X %8d\n",
        i, prptr->pname, pstate[(int)prptr->pstate],
        prptr->pprio, prptr->pparent, prptr->prstkbase,
        prptr->prstkptr, prptr->prstklen);
}

return 0;
}
```

- Hasilnya

```

}

/* Print header for items from the process table */

printf("%3s %16s %5s %4s %4s %10s %10s\n",
       "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
       "Stack Ptr", "Stack Size");

printf("%3s %16s %5s %4s %4s %10s %10s\n",
       "----", "-----", "-----", "----", "-----",
       "-----", "-----");

/* Output information for each process */
/* Rengganis Tantri Pramudita, 2311104065, SE0702 */

for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %16s %5s %4d %4d 0x%08X 0x%08X %8d\n",
           i, prptr->pname, pstate[(int)prptr->prstate],
           prptr->prprio, prptr->prparent, prptr->prstkbase,
           prptr->prstkptr, prptr->prstklen);
}
return 0;
}
xinu@xinu-develop-end:~/xinu$

```

- Ubahlah perintah ps (source code: xsh_ps.c) pada Xinu sehingga menampilkan informasi tambahan berupa kolom yang berisi total message yang ada pada proses seperti gambar di bawah ini:

```
xsh $ ps
```

Pid	Name	State	Prio	Ppid	Stack Base	Stack Ptr	Stack Size	Msg	Content
0	prnull	ready	0	0	0x005FDFFC	0x00FFFF04	8192	1	4
1	rdsproc	susp	200	0	0x005FBFFC	0x005FBFC8	16384	0	0
2	ipout	wait	500	0	0x005F7FFC	0x005F7F40	8192	0	0
3	netin	wait	500	0	0x005F5FFC	0x005F5EE0	8192	0	0
5	Main process	recv	20	4	0x005E3FFC	0x005E3F64	65536	0	0
6	shell	recv	50	5	0x005D3FFC	0x005D3C7C	8192	0	0
7	ps	curr	20	6	0x005F3FFC	0x005F3DC8	8192	0	0

Kolom Msg adalah banyaknya pesan yang ada dalam proses.

Kolom Content adalah isi dari pesan tersebut.

Langkah pengerjaan:

- Modifikasi source code pada file xsh_ps.c

Jawab:

- Sebelum

```

File Edit View Search Terminal Help
/* xsh_ps.c - xsh_ps */

#include <inu.h>
#include <stdio.h>
#include <string.h>

/*-----
 * xsh_ps - shell command to print the process table
 *-----
 */
shellcmd xsh_ps(int nargs, char *args[])
{
    struct procent *prptr;          /* Pointer to process */
    int32 i;                        /* Index into proctabl */
    char *pstate[] = {              /* Names for process states */
        "free ", "curr ", "ready", "recv ", "sleep", "susp ",
        "wait ", "rttime");

    /* For argument '--help', emit help about the 'ps' command */

    if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
        printf("Use: %s\n", args[0]);
        printf("Description:\n");
        printf("\tDisplays information about running processes\n");
        printf("Options:\n");
        printf("\t--help\t display this help and exit\n");
        return 0;
    }
}

```

```

File Edit View Search Terminal Help
}

/* Check for valid number of arguments */

if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
        args[0]);
    return 1;
}

/* Print header for items from the process table */

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
    "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
    "Stack Ptr", "Stack Size");

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
    "....", "-----", "-----", "-----", "-----",
    "-----", "-----", "-----");

/* Output information for each process */
/* Rengganis Tantri Pramudita, 2311104065, SE0702 */

for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->pstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
        i, prptr->pname, pstate[(int)prptr->pstate],
        prptr->prprio, prptr->prparent, prptr->prstkbase,
        prptr->prstkptr, prptr->prstklen);
}
return 0;
}

```

- Sesudah modifikasi

```

/* Print header for items from the process table */

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
       "Pid", "tName", "State", "Prio", "Ppid", "Stack Base",
       "Stack Ptr", "Stack Size", "Msg", "Content");

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
       "----", "-----", "-----", "-----", "-----", "-----",
       "-----", "-----", "-----", "-----");

/* Output information for each process */
/* Rengganis Tantri Pramudita, 2311104065, SE0702 */

for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
           i, prptr->prname, prptr->prstate, prptr->prprio, prptr->pparent, prptr->prstkbase,
           prptr->prstkptr, prptr->prstklen, prptr->prhasmsg, prptr->prmsg);
};
    }
    return 0;
}

```

- Jalankan Backend VM
- Setelah sistem berjalan, jalankan perintah \$ps. Pastikan hasilnya sesuai dengan contoh output pada gambar yang diberikan.
- Screenshot source kode dan output akhir hasil modifikasi

Jawab:

```

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
 0 prnull         ready  0    0 0x005FDFFC 0x00FFFE90   8192  1    3
 1 ipout          wait  500   0 0x005FBFFC 0x005FBF30   8192  0    0
 2 netin          wait  500   0 0x005F9FFC 0x005F9ED0   8192  0    0
 4 Main process   recv  20   3 0x005E7FFC 0x005E7F34  65536  0    0
 5 shell          recv  50   4 0x005D7FFC 0x005D79AC   8192  0    0
 6 ps             curr  20   5 0x005F7FFC 0x005F7DD8   8192  0    0
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0

```

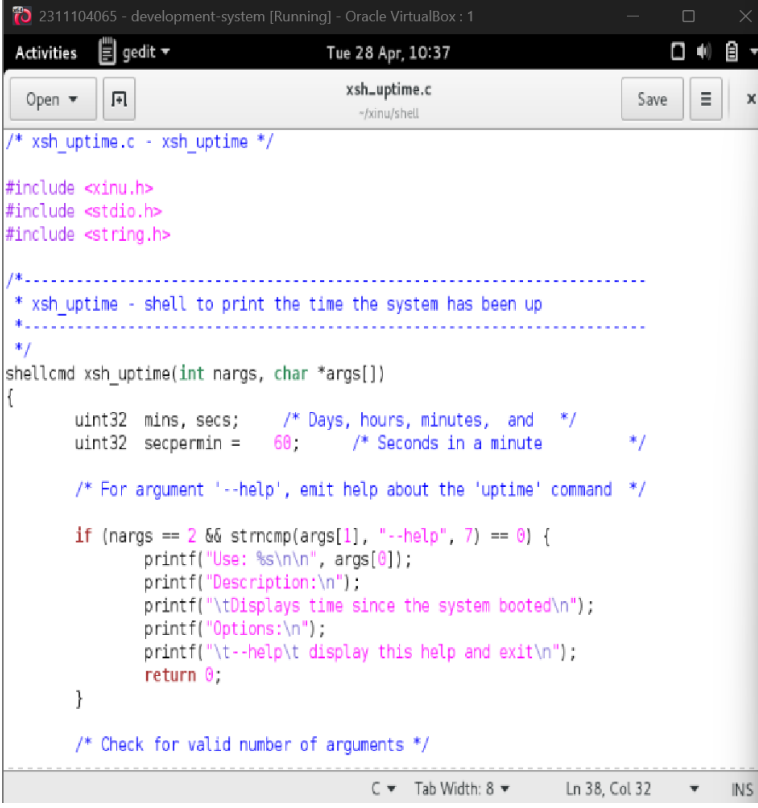
4. Ubahlah perintah uptime pada Xinu sehingga menampilkan lamanya Xinu sejak booting hanya dalam satuan menit.

Langkah pengerjaan:

- Modifikasi source code pada file xsh_uptime.c

Jawab:

Setelah dimodifikasi



The screenshot shows a gedit editor window titled 'xsh_uptime.c' with the path '~/.xinu/shell'. The code is as follows:

```
/* xsh_uptime.c - xsh_uptime */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

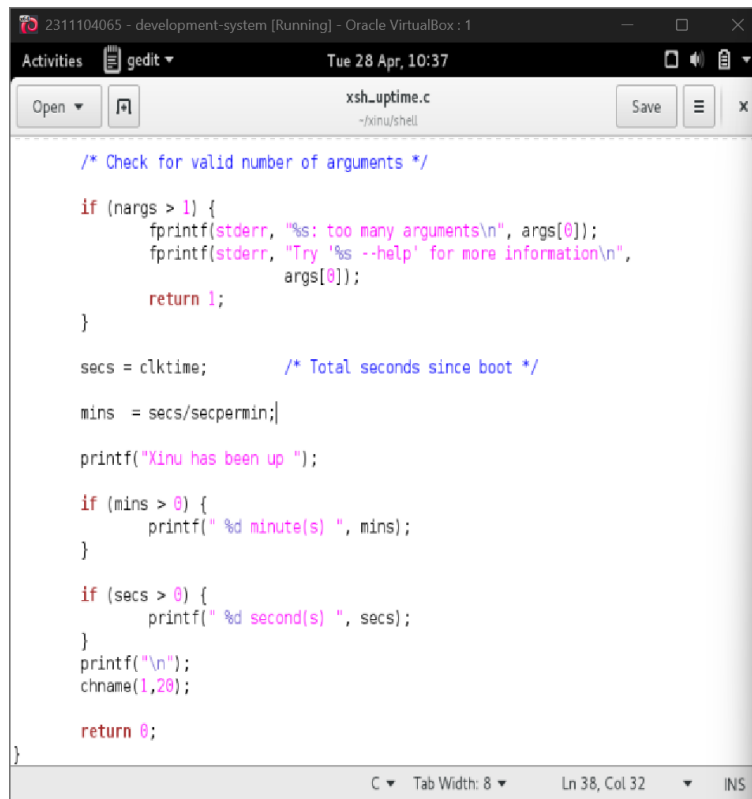
/*-----
 * xsh_uptime - shell to print the time the system has been up
 *-----
 */
shellcmd xsh_uptime(int nargs, char *args[])
{
    uint32 mins, secs;    /* Days, hours, minutes, and */
    uint32 secpersmin = 60; /* Seconds in a minute */

    /* For argument '--help', emit help about the 'uptime' command */

    if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
        printf("Use: %s\n\n", args[0]);
        printf("Description:\n");
        printf("\tDisplays time since the system booted\n");
        printf("Options:\n");
        printf("\t--help\t display this help and exit\n");
        return 0;
    }

    /* Check for valid number of arguments */
```

The status bar at the bottom indicates 'C', 'Tab Width: 8', 'Ln 38, Col 32', and 'INS'.



```
/* Check for valid number of arguments */
if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

secs = clktime;      /* Total seconds since boot */
mins = secs/secpermin;

printf("Xinu has been up ");

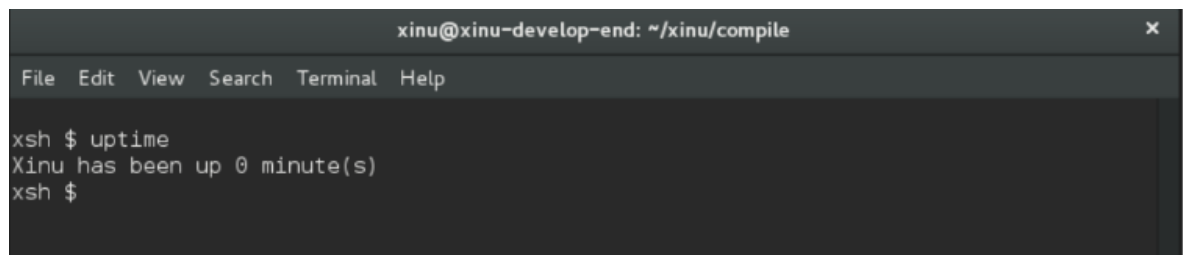
if (mins > 0) {
    printf(" %d minute(s) ", mins);
}

if (secs > 0) {
    printf(" %d second(s) ", secs);
}
printf("\n");
chname(1,20);

return 0;
}
```

- Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya
- Jalankan Backend VM
- Setelah sistem berjalan, jalankan perintah \$uptime. Pastikan hasilnya sesuai dengan contoh output yang diinginkan
- Screenshot source kode dan output akhir hasil modifikasi

Jawab:



```
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

xsh $ uptime
Xinu has been up 0 minute(s)
xsh $
```